BENJAMIN KOFI YANKEY

Full-Stack Software Developer | Python · TypeScript/Node.js · Java · C++
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PROFESSIONAL SUMMARY

Full-stack developer and DevOps engineer with hands-on experience writing, debugging, and reviewing production code in Python, TypeScript/JavaScript (Node.js), Java, and C++/C#, backed by a Computer Science degree with strong grounding in data structures and algorithms. Has independently built and shipped full-stack applications end-to-end — from React/Angular front-ends and Express, Flask and FastAPI back-ends to PostgreSQL, Redis and DynamoDB data layers — and deployed them on AWS with Terraform, containers and serverless services. Regularly works across unfamiliar codebases, tracks down root causes of bugs and security/quality issues using automated tooling, and documents technical reasoning clearly for teammates — well suited to constructing clean, reproducible reference solutions and explaining the reasoning behind them.

RELEVANT QUALIFICATIONS

- **Core Languages:** 1+ years hands-on with Python and JavaScript/TypeScript (Node.js); academic and project experience in Java, C++, and C# across full-stack and systems-oriented coursework.
- **Algorithms & Data Structures:** Solid foundation from a Computer Science degree — arrays, linked lists, trees, graphs, sorting/searching, dynamic programming — applied in coursework and production systems work.
- **Building Features End-to-End:** Designed and shipped complete applications including a customer-credit tracking app for a retail shop (React, TypeScript, Express, Prisma/PostgreSQL), a microservices e-commerce storefront with JWT auth and an admin panel, and a serverless task manager with role-based access control.
- **Debugging & Root-Cause Analysis:** Integrated and acted on findings from Trivy, Gitleaks, OWASP Dependency Check, and SonarQube, catching and resolving dozens of vulnerabilities and code-quality issues before deployment.
- **Testing & Reliability:** Wrote automated test suites (pytest, Jest) run as pipeline gates, and built a self-healing monitoring stack that detects failing services and remediates them automatically.
- **Working Across Codebases:** Navigated large, complex, unfamiliar codebases independently as a trainee developer, supplemented by focused self-study and mentorship from senior engineers.
- **Documentation & Communication:** Authored detailed READMEs, architecture docs and repeatable, auditable infrastructure (Terraform/CloudFormation) so that reasoning and configuration decisions are clear to the rest of the team.
- **Peer Code Review:** Actively participated in team code reviews, providing constructive, specific feedback to uphold code quality and reinforce best practices.

PROFESSIONAL EXPERIENCE

DevOps Engineer
Nov. 2025 – Present

Amalitech | Takoradi, Ghana

- Established and maintained end-to-end CI/CD pipelines (Jenkins, GitHub Actions) covering build, test, and deployment, with blue-green/rolling strategies and rollback mechanisms, cutting average release cycle times by 35%.
- Integrated automated security and quality scanning (Trivy, Gitleaks, OWASP Dependency Check, SonarQube) into CI/CD, catching dozens of vulnerabilities and code-quality issues before deployment.
- Containerized 10+ applications with Docker and orchestrated them on Kubernetes (Deployments, Services, ConfigMaps, Secrets, HPA), debugging environment and configuration issues to ensure consistency across environments.
- Architected and documented production-grade AWS infrastructure with Terraform and CloudFormation across 10+ services, keeping configuration version-controlled, auditable, and reproducible.
- Configured monitoring and logging (CloudWatch, Prometheus, Grafana), improving system visibility, uptime by 25%, and incident response time.

AWS re/Start Intern
Sept. 2025 – Nov. 2025

Amalitech | Kumasi, Ghana

- Tuned and scaled Amazon RDS databases and integrated S3 with EC2 via instance profiles and pre-signed URLs, improving query efficiency by 40%.
- Provisioned and configured 10+ AWS services (EC2, S3, IAM, CloudFormation) for repeatable infrastructure deployment across multiple environments.

- Engineered secure VPC architectures with public/private subnets and least-privilege IAM, reducing security misconfigurations by 30% per AWS Trusted Advisor findings.

Trainee Software Developer

Oct. 2022 – Nov. 2022

Amalitech | Takoradi, Ghana

- Contributed to multiple features across full-stack web applications as part of a 5–10 person engineering team, using JavaScript, HTML, CSS, and React to connect front-end interfaces to back-end data sources.
- Actively participated in team code reviews, providing constructive feedback to peers to uphold code quality and adherence to best practices.
- Applied Git and GitHub for version control and collaboration across the team's shared codebases.
- Strengthened analytical and problem-solving abilities by navigating large, complex codebases independently, supplemented by focused self-study and mentorship from senior colleagues.

SELECTED PROJECTS

Software Engineering

Thrift Hub — Customer & Credit Tracking App *(React, Vite, TypeScript, Tailwind CSS, Node.js/Express, PostgreSQL, Prisma)* github.com/Benjamin-yankey/Hub

Built and shipped a working web app for a retail shop that registers customers with permanent IDs, records purchases and payments, and automatically recalculates outstanding balances per purchase and per customer. Includes customer search with full purchase history, a dashboard of debtors, daily sales for till reconciliation and a 7-day trend, and logging of walk-in cash by payment method (cash / mobile money).

ShopNow — Microservices E-Commerce Platform *(Angular, Node.js/Express, PostgreSQL, Redis, Docker, Terraform, ECS Fargate, EKS, Helm)* github.com/Benjamin-yankey/Poly-Orchestrator

Developed a full storefront — catalog search and filtering, JWT-secured accounts, per-user carts, checkout via a mock payment gateway, order history, and a role-gated admin panel for product CRUD — using a database-per-service design. Deployed the same container images to both ECS Fargate and EKS from a single Terraform configuration with CI/CD, to benchmark the two orchestrators.

Serverless Task Management System *(React, TypeScript, AWS Lambda (Node.js), API Gateway, Cognito, DynamoDB, SNS/SES, Amplify, Terraform)* github.com/Benjamin-yankey/serverless-task-management

Built a task manager with admin and member roles (create, assign, prioritise and close tasks), Cognito-authorized APIs, email-domain restrictions, Google/GitHub OAuth, and email notifications on assignment; responsive, accessible UI with live search and filterable stats.

DevLog — Developer Blogging Platform *(React, Node.js, PostgreSQL, Docker, Jenkins)* github.com/Benjamin-yankey/commentary-blog

Developed a publishing platform with a markdown-capable editor, auto-save, tagging, nested comments, author follows and full-text search, secured with JWT and bcrypt; packaged with multi-stage Docker builds and a Jenkins CI pipeline.

MediCo — Telemedicine Web Application *(React, Node.js)*

Built a full-stack prototype spanning virtual consultations, secure lab report management, and home care scheduling; structured the data layer for secure storage and retrieval of patient records.

Cloud, DevOps & Reliability

Self-Healing Monitoring System *(Python/Flask, PostgreSQL, Redis, Prometheus, Grafana, Alertmanager, Terraform, Lambda, SSM, EventBridge)* github.com/Benjamin-yankey/AI-Devops

Built a multi-tier project/task manager (REST API, background job worker, Postgres, Redis cache/queue) instrumented with golden-signal metrics; alert rules trigger automatic remediation — restarting the API when the 5xx rate exceeds 5% or the worker when the queue backlog passes 50 jobs — to reduce MTTR. Includes a local Docker Compose version and an AWS version.

FinCorp — Immutable Supply Chain & Disaster Recovery *(FastAPI, pytest, CodeArtifact, CodePipeline, CodeBuild, ECR, RDS PostgreSQL, AWS Backup, Terraform)* github.com/Benjamin-yankey/Artifacts

Built a FastAPI double-entry ledger service with a pytest suite and an AWS-native pipeline that tests, scans and blocks images with HIGH/CRITICAL vulnerabilities, using immutable ECR tags; designed cross-region backups to restore the database in a second region in under 30 minutes.

GitOps Mail System — Secure Delivery to ECS *(Node.js/Express, Jenkins, SonarQube, Gitleaks, Trivy, Terraform, ECS Fargate)*
github.com/Benjamin-yankey/Gitops_lab

Built a mail web app (folders, search, live unread/starred counts) delivered through a 19-stage Jenkins pipeline with SAST, dependency, secret and container scanning gates before deployment to ECS Fargate.

Achievement & Utilization Tracking System *(Terraform, Jenkins, Docker, ECS, ECR)*

Assembled an end-to-end CI/CD pipeline for full-stack apps using reusable Terraform modules; integrated SonarQube, Trivy, and Gitleaks scanning.

Node.js App Deployment & Observability on AWS *(EC2, Docker, Nginx, PM2, Jenkins, Terraform, Prometheus, Grafana, Jaeger)*
github.com/Benjamin-yankey/Logging-and-monitoring

Deployed a production Node.js app with Jest tests, hardened security headers, rate limiting and input validation, behind an Nginx reverse proxy, with no unplanned outages during the deployment period; added metrics, tracing, dashboards and alerting.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript (Node.js), Java, C++, C, C#, SQL, Bash

Full-Stack Development: React, Angular, Vite, Tailwind CSS, Node.js/Express, Flask, FastAPI, Prisma, REST APIs, JWT/OAuth, HTML/CSS, OOP & functional paradigms

Databases: PostgreSQL, MySQL, Redis, DynamoDB, Amazon RDS

Testing & Tooling: pytest, Jest, Postman, Git/GitHub, Agile/Scrum

Data Structures & Algorithms: Arrays, linked lists, trees, graphs, sorting/searching, dynamic programming

Cloud & IaC: AWS (EC2, ECS/Fargate, EKS, ECR, Lambda, API Gateway, Cognito, DynamoDB, RDS, S3, IAM, VPC, CloudFormation, CloudWatch, EventBridge, SNS/SES, Route 53, CodePipeline, CodeBuild, AWS Backup), Terraform, Ansible

Containers & CI/CD: Docker, Docker Compose, Kubernetes, Helm, Jenkins, GitHub Actions, GitOps, blue-green/rolling deployments

Code Quality & Security: Trivy, Gitleaks, OWASP Dependency Check, SonarQube, IAM policies, TLS/SSL, RBAC

Monitoring: CloudWatch, Prometheus, Grafana, Alertmanager, Jaeger, ELK Stack, structured log aggregation, alerting

EDUCATION & CERTIFICATIONS

B.Sc. Computer Science

Sept. 2021 – Aug. 2025

Kwame Nkrumah University of Science and Technology (KNUST) | Kumasi, Ghana

- Relevant coursework: Data Structures & Algorithms, Databases & Data Management Systems, Software Engineering, Operating Systems, Computer Networks, Distributed Systems, Cybersecurity Fundamentals, Machine Learning & AI.

Certifications

- **AWS Certified Cloud Practitioner** — Amazon Web Services, January 2025 View certificate
- **AWS re/Start Graduate** — Amalitech / Amazon Web Services, 2025
- **AWS Course Completion Certificate** — Amazon Web Services View certificate
- **AWS Course Completion Certificate** — Amazon Web Services View certificate
- **Amazon S3** — Course Certificate View certificate
- **Data Migration** — Course Certificate View certificate
- **NDG Linux Essentials** — Cisco Networking Academy / Network Development Group View certificate
- **Software Engineer Certificate** — IPMC Ghana
- **HTML and CSS Certificate** — freeCodeCamp
- **Additional certificates:** Certificate 1 · Certificate 2 · Certificate 3